

The Impact of Information Systems in High-Stakes Decisions: Evidence from Physicians *

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Abstract

How do professionals respond to computerized, data-driven guidance in practice? We analyze a workers' compensation insurance program where physicians make coverage and diagnosis decisions. We study the introduction of an automated system that flagged diagnoses with historically low coverage. We develop a model that yields testable predictions to distinguish between informational and persuasive effects. Consistent with persuasion, physicians granted coverage less often on alert-triggering diagnoses, but physicians also avoided those alerts by recoding diagnoses. Finally, secondary reviews show the system aligned physician decisions with intended coverage standards. These findings provide lessons for the design of information systems for decision-makers.

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1 Introduction

Bankers, teachers, judges, physicians, and other experts make decisions that carry significant implications for others. Indeed, loan approvals, high-stakes examinations, petitions for asylum, and medical diagnoses can all influence individuals’ outcomes in substantial and enduring ways. Yet, all decision-makers can make mistakes or be heavily influenced by unrelated environmental or social factors—such as air pollution or even the outcome of a sports event.¹

The growing adoption of artificial intelligence and machine learning in recent years has raised important questions about how professionals interact with information systems and whether such systems improve decision accuracy. Because prediction algorithms rely on historical and partial data, and given the associated ethical considerations, most systems do not operate autonomously (McLaughlin and Spiess, 2024). Instead, they generate information or statistical assessments that decision-makers may incorporate into their judgments (Dietvorst et al., 2018; Kawaguchi, 2021; Stevenson and Doleac, 2024). However, there is limited evidence on how professionals respond to such systems in the field, particularly in high-stakes contexts, despite their importance for evaluating effectiveness and informing optimal design.²

This study examines how professionals adjust their decisions in response to data-driven information in the field. We focus on coverage decisions within the largest

¹For example, ingroup biases have been documented among judges (e.g., Shayo and Zussman, 2011; Kuran and Lustig, 2012), teachers (Hanna and Linden, 2012; Lavy et al., 2022), and sports referees (Price and Wolfers, 2010; Parsons et al., 2011). Moreover, judicial decisions have been found to vary with irrelevant factors such as the defendant’s birthday (Chen and Philippe, 2023) or unexpected losses of local football teams (Eren and Mocan, 2018). Past decisions also bias judges, loan reviewers, and baseball umpires (Chen et al., 2016). There is growing evidence that air pollution impairs cognitive performance and productivity among investors and students (Huang et al., 2020; Ebenstein et al., 2016). Finally, Rambachan (2024) examines systematic judicial errors and argues that algorithmic decision-making may improve outcomes.

²Angelova et al. (2026) is a notable recent exception studying algorithmic recommendations in bail decisions.

provider of workers' compensation insurance in Chile, which covers approximately 40 percent of salaried employees in the country. In this setting, primary care physicians not only deliver medical care but also determine whether patients' conditions stem from work-related incidents and thus qualify for financial coverage. Because the law specifies eligibility criteria only in broad terms, physicians retain substantial discretion in these decisions. This discretion creates an agency problem: the insurer has a financial interest in potentially limiting coverage, while physicians may weigh patient welfare or their own assessment of the merits of the case.

To provide empirical evidence on how data-driven information shapes decisions in the field, we exploit the introduction of a computerized alert system in a subset of clinics within the insurer's network. The system flagged diagnoses that historically had low coverage rates by displaying a pop-up notification on the physicians' computers after they had entered selected diagnosis codes. The notification stated that the selected diagnosis was likely to be ineligible for coverage. Importantly, physicians have discretion on the coded diagnosis, and thus may strategically adjust diagnosis choices to avoid triggering alerts. We exploit this feature of the setting in two key ways. First, it enables us to uncover the mechanism through which the message operates. Second, it allows us to examine not only how professionals respond to algorithmic information but also how they may influence the algorithm's output and, in turn, its effectiveness.

We develop a model of physician behavior to examine how this data-driven intervention can influence coverage and coding decisions and to guide our empirical analysis. In the model, physicians determine both whether to grant coverage and which diagnosis to code. We introduce two key mechanisms through which the alert system may affect behavior. First, the alert may provide physicians with *new information* relevant to assessing the merits of each case. Second, although physicians derive in-

trinsic utility from granting coverage because they internalize patients’ welfare (Dickstein, 2011; Iizuka, 2012; Schnell, 2017), the alert can increase the perceived cost of deviating from the implicit recommendation in the message (McLaughlin and Spiess, 2022). This feature generates a *persuasion* mechanism, whereby the alert influences physicians independently of its informational content (DellaVigna and Gentzkow, 2010). Physicians may also respond strategically by engaging in costly “recoding”—modifying diagnosis codes to avoid triggering an alert (Dafny, 2005; Geruso and Layton, 2020). When recoding costs are low, this behavior can limit the effectiveness of persuasion. Our framework highlights recoding as a novel mechanism shaping how decision-makers respond to algorithm-based information systems.

The model yields testable empirical predictions. Under an information-based mechanism, additional information on past coverage rates should not induce recoding, as it does not alter the medical diagnosis. Moreover, information effects should extend to both alert and non-alert cases and be stronger among less-experienced physicians. In contrast, a persuasion mechanism predicts strategic recoding, with the intensity of recoding depending on the costs of doing so. It also implies coverage effects concentrated on alert-triggering cases and greater alignment between physicians’ decisions and organizational preferences. Thus, because physicians can recode diagnoses to avoid the alert, and because information alone gives no reason to do so, recoding reveals a compliance cost that operates independently of the merits of the case. These predictions inform our empirical strategy for identifying the mechanisms underlying physician responses to the computerized alert system.

We test the model predictions by estimating the alert system’s effect on physicians’ diagnosis coding and coverage decisions. We also examine coverage reversal decisions by review committees in every clinic. To this end, we implement a difference-in-differences design that compares changes in these outcomes before and after system

implementation between participating and non-participating clinics and investigates relevant margins of heterogeneity. To address the endogeneity arising from strategic recoding, we also use an instrumental-variable approach based on features of the institutional setting.

We find that the alert system induced significant recoding behavior: the system reduced physicians' likelihood of coding an alert-triggering diagnosis by 7 percent. Recoding was particularly prevalent when physicians could substitute toward similar diagnoses that did not trigger the alert. Beyond recoding, coverage decreases when an alert is triggered. Physicians are 13 percentage points less likely to grant coverage when an alert appears—a 65 percent reduction relative to the 20 percent coverage rate among alert-triggering diagnoses.

Our findings suggest that the alert system affects physician behavior primarily through persuasion rather than through information. The presence of recoding suggests that physicians have an intrinsic preference for granting coverage, and that the alert increased the perceived cost of doing so. Two additional findings are inconsistent with an information-based mechanism being at play. First, we find null treatment effects of the alert on coverage rates for diagnoses that did not trigger an alert. If the system operated through information transmission, the absence of an alert should also convey meaningful information—specifically, that the historical coverage rate is *not* low—;yet we see no behavioral response to such a signal. Second, we find no evidence that more experienced physicians respond less strongly to the alert than less experienced ones. This runs counter to the prediction that better-informed physicians should rely less on informative messages, as would be expected under an information-based mechanism.

We also leverage the fact that a committee of secondary reviewers may overturn the physicians' coverage decisions. The alert reduced the reversal rate by 24 percent,

showing that the marginal cases denied coverage under the alert had weaker merits than the inframarginal cases that continued to receive coverage.

Our results provide important insights into how decision-makers interact with information systems in the field. These insights are even more important with the rise of more intelligent recommendation systems. We show that computerized interventions can influence decisions through a preference-based mechanism, particularly when they impose a (perceived) cost of non-compliance. At the same time, when decisions depend on factors beyond information, professionals can offset the intended effects of such interventions if the cost of doing so is low. These findings underscore the importance of understanding decision-makers' preferences when designing data-driven interventions to influence expert behavior in the field.

The persuasion effect also implies that professionals may follow similar types of alerts even against their professional judgment. This effect is likely to be particularly prevalent when algorithms are deployed in high-stakes environments. In such contexts, a mistake made while following an implicit recommendation is less likely to be attributable to the individual than a mistake that occurred while deciding against the recommendation. The liability environment is likely to make this mechanism particularly salient in health care or similar contexts.

This article contributes to a growing strand of the literature that examines the effects of information technology (IT) adoption in organizations. In health care, Athey and Stern (2002), Agha (2014), and McCullough et al. (2016) evaluate the effect of IT adoption on patient outcomes, while Epstein and Ketcham (2014) studies the impact of IT on physicians' prescribing decisions.³ Our contribution lies in

³Bundorf et al. (2019) analyzes the effect of an artificial intelligence recommendation system on health insurance choice (Medicare Part D). Also, a broad set of papers study the effects of adoption of electronic medical records (Parente and McCullough, 2009; McCullough et al., 2010; Miller and Tucker, 2011) or clinical decision support systems (Ganju et al., 2020) on patient outcomes, hospital quality, and discrimination.

studying the real-world implementation of computerized information provision, addressing a gap in the literature that has predominantly relied on simulations (Ribers and Ullrich, 2019; Hastings et al., 2020).⁴ Agarwal et al. (2024) develops a randomized controlled trial to study the effects of providing AI-based predictions on radiologists’ retrospective screening for potential chest pathologies. Their findings highlight that additional information does not automatically improve decisions when decision-makers are biased. We complement their work by providing field evidence showing that decision-makers exhibit preferences for complying with implicit recommendations from information systems.

Some recent studies analyze the potential role of data-driven methods in assisting decision-making. Kawaguchi (2021) analyzes conditions affecting retail workers’ adoption of algorithmic recommendations, while Lebovitz et al. (2022) found that radiologists abandoned machine learning recommendations after initial use. Sun et al. (2022) show that employees significantly deviated from algorithmic recommendations. Laboratory studies by Dietvorst et al. (2015, 2018) reveal that recommendation uptake increases when decision-makers can modify algorithms themselves, addressing the so-called “algorithmic aversion”. We study whether physicians complied with the recommendations provided by the organization’s computer system in a setting that has large stakes for patients. We find that a simple message conveying aggregate information can have substantial impacts on high-stakes decisions at the margin.

This article also contributes to a vast empirical literature that analyzes the effects of communication, which seeks to quantify the effects of communication on behav-

⁴A recent exception is Goh et al. (2025), which examines the effect of giving physicians the option of interacting with a large language model. In contrast, we study information delivery, which is very common. Applications include media platforms (Hallinan and Striphas, 2016), food labels (Alé-Chilet and Moshary, 2020), quality certification of restaurants (Dai et al., 2018), and health insurance plans (McCarthy and Darden, 2017).

ior and to understand the mechanisms through which it operates (Gentzkow and Shapiro, 2006; DellaVigna and Kaplan, 2007; Martin and Yurukoglu, 2017; Dubois and Tunçel, 2021). Our setting also allows us to examine the mechanism through which the message acted. We distinguish between information- and preference-based models (Akerberg, 2001; DellaVigna and Gentzkow, 2010). Our results show that decision-makers react to these messages by changing preferences, and not because of the specific information they contain.⁵

Our empirical context is workers’ compensation insurance (Krueger and Burton Jr, 1990; Gruber and Krueger, 1991; Cabral et al., 2022). We complement concurrent work by Cabral and Dillender (2025) studying heterogeneity in coverage generosity in a different setting. While their analysis considers counterfactual policies that reallocate patients across physicians with fixed generosity, we evaluate the effect of an implemented policy designed to alter physicians’ generosity without changing the nature of the patient-physician matching process.

2 Institutional Details

Chilean law mandates that all employers carry workers’ compensation insurance, which covers work-related injuries and illnesses.⁶ This insurance is provided either by a public insurer or by one of three non-profit private organizations. Our partner organization, which we call INS henceforth, is the largest private insurer in the country. INS operates a vertically integrated system comprising 89 outpatient clinics and

⁵Relatedly, Xiang (2026) studies a setting where a partially altruistic, informed physician designs a recommendation strategy to steer a patient toward more profitable treatment. Our setting reverses the role of persuasion: the physician is the receiver of an institutionally designed signal rather than its sender.

⁶Workers’ compensation covers incidents including injuries sustained while performing job duties and accidents occurring during commutes to and from the workplace. Coverage criteria are defined by law. All other health-related events fall under the general health system.

hospitals nationwide. Private insurers collectively cover approximately 80 percent of Chile’s workforce, with INS serving about half of that population, totaling more than 2.6 million workers.⁷

By law, workers’ compensation insurance offers comprehensive benefits, including access to the insurer’s high-quality provider network, full coverage of medical expenses (such as physical and occupational therapy and transportation to and from health centers), pensions for permanent disability (if applicable), and full salary replacement during medical leave.⁸ In contrast, the general health system often requires copayments and does not provide wage compensation for the first three days of sick leave lasting 10 days or less.⁹

Patients who visit an INS clinic are evaluated by a primary care physician (PCP), who provides initial medical treatment and assesses whether the case qualifies for coverage under statutory guidelines. The law specifies only broad eligibility criteria, which depend on the circumstances of the incident, the employee’s preexisting health conditions, and the nature of their occupational duties.

After the PCP’s initial coverage determination, a clinic-based committee may review the case and overturn the decision. These review committees consist of the clinic’s head physician, administrative manager, and an assistant.¹⁰

2.1 The Alert Experiment

Our partner organization implemented an intervention aimed at standardizing coverage decisions, motivated by the lack of specific eligibility guidelines and high physi-

⁷Statistics of Social Security, 2023: <https://www.suseso.cl/608/w3-propertyvalue-10364.html>. Accessed January 16, 2025.

⁸Surveys report that 77% of workers were satisfied or very satisfied with these services (“Encuesta SUSESO 2023,” Fibra Research, November 2023, https://www.suseso.cl/607/articles-724468_archivo_01.pdf).

⁹Approximately 40% of sick leaves in Chile are shorter than 10 days (Celhay et al., 2024).

¹⁰See additional institutional details in Appendix Section A.

cian turnover.¹¹ The intervention consisted of an automated pop-up alert integrated into the PCP’s computer interface. These alerts appeared when physicians entered a diagnosis code within a pre-specified list of diagnoses with low historical coverage rates.¹² The alert pop-up read:

*“According to historical decisions of physicians in INS, the chosen diagnosis has a high probability of being ineligible for coverage.”*¹³

To evaluate the effectiveness of the alert system, the insurer initially implemented it as a pilot program between November 8, 2019, and February 17, 2020, in 25 of the network’s 89 clinics. Following the pilot, the insurer expanded the system to all clinics. The pilot clinics were selected to be representative of the network as a whole.¹⁴

3 Data and Descriptive Statistics

We use administrative data covering all cases reviewed for coverage in INS clinics from January 1, 2018, to February 16, 2020, the date when the alert system was expanded to all clinics. The records include physicians’ coverage decisions, any reversal decisions by review committees, patient demographics (gender and age), clinic and physician identifiers, and physician characteristics (gender, age, and tenure at INS). The data also contain internal diagnosis codes (5,357 possible codes, with 2,184

¹¹The median physician tenure at INS is 24 months, with approximately 50 percent of physicians leaving within 12 months in our data. Physicians tend to be young, with a median age of 33 years, and many leave INS to pursue medical specialization.

¹²The diagnoses triggering an alert were selected based on coverage data from January 2018 to June 2019. Appendix B reports the most frequent alert- and non-alert-triggering diagnoses. Some alert-triggering diagnoses (e.g., mild finger contusion) have closely related, more severe counterparts that did not trigger alerts (e.g., mild finger wound).

¹³Translation is our own. Appendix C shows a computer screenshot of the alert.

¹⁴In Section 3, we show that pilot clinics do not differ significantly from non-pilot clinics along key dimensions.

unique codes observed), which are grouped into 52 diagnosis categories encompassing clinically similar conditions. In the pre-intervention period, there were on average 9,990 visits per month, 134 visits per clinic, and between 1 and 5 unique active physicians per clinic (10th to 90th percentiles).¹⁵

We apply two sample restrictions. First, we exclude 577 physicians with fewer than 10 visits pre-implementation or 5 visits post-implementation. Second, to reduce the possibility of treatment contamination, we drop 56 physicians who spent less than 95% of their time in either treated or control clinics during the experimental period. The final sample includes 259 physicians, 238,280 patients, and 262,504 cases.

Table 1: Summary Statistics

	All Mean	Control clinics Mean	Treated clinics Mean	<i>p</i> -value
A. Patient Characteristics by Clinic				
Female (%)	39.2	38.7	40.3	0.41
Age	40.2	40.2	40.1	0.58
No. patients	2,503.9	1,928.0	3,817.0	0.20
B. Physician Characteristics by Clinic				
Female (%)	27.2	29.2	22.6	0.28
Age	35.4	35.3	35.6	0.89
No. Physicians	10.2	8.7	13.7	0.27
C. Visit Outcomes by Clinic				
Alert (%)	14.7	14.7	14.5	0.88
Coverage (%)	81.3	81.2	81.5	0.83
Correction to no coverage (%)	4.7	4.6	5.0	0.35
Correction to coverage (%)	0.3	0.4	0.3	0.49
No. Cases	2,680.3	2,085.9	4,035.6	0.21
No. Clinics	82	57	25	

Note: Descriptive statistics for the pre-intervention period. An observation in any of the three panels is a clinic. *p*-values indicate statistical differences between control and treatment clinics.

Table 1 reports descriptive statistics, with Panels A, B and C presenting clinic

¹⁵We do not have data on the full alert log or on the initial diagnosis code, in case it was changed by the physician after an alert.

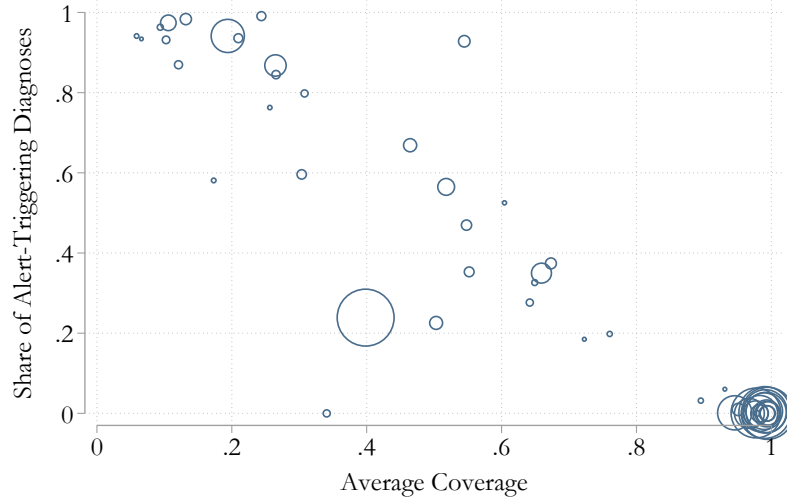
averages for patients’, physicians’, and visits’ characteristics, respectively. Column (1) shows averages for the set of clinics, while Columns (2) and (3) report averages separately for treated and control clinics. Treated and control clinics differ on some characteristics, most visibly clinic size. Yet, none of the differences between treated and control clinics are statistically significant. Because the Table aggregates visits to the clinic level, those tests have limited power to detect compositional differences. Appendix D reports visit-level (and clinic-month-level) balance regressions of pre-determined patient and physician characteristics on treatment status and its interaction with the post period. The interaction coefficient is small and not statistically significant for all characteristics, both with and without clinic fixed effects. We do observe level differences in that treated clinics employ physicians with shorter average tenure, which are absorbed by the clinic or physician fixed effects in the empirical analysis.¹⁶

Pre-intervention coverage rates vary across diagnosis groups and are, as expected, negatively correlated with the share of alert-triggering diagnoses within each group: 92 percent of visits with a non-alert triggering diagnosis code are given coverage pre-intervention, while this number decreases to 20 percent for alert-triggering diagnoses. Figure 1 plots the share of alert-triggering diagnoses and the average coverage rate during the pre-intervention period for each diagnosis group. The negative slope reflects that alert-triggering diagnoses were selected from those with the lowest historical coverage.

As in Cabral and Dillender (2025), we find that physicians exhibit substantial heterogeneity in coverage decisions. Figure 2 plots the density of physicians’ relative coverage propensities, both unconditional and conditional on observable visit charac-

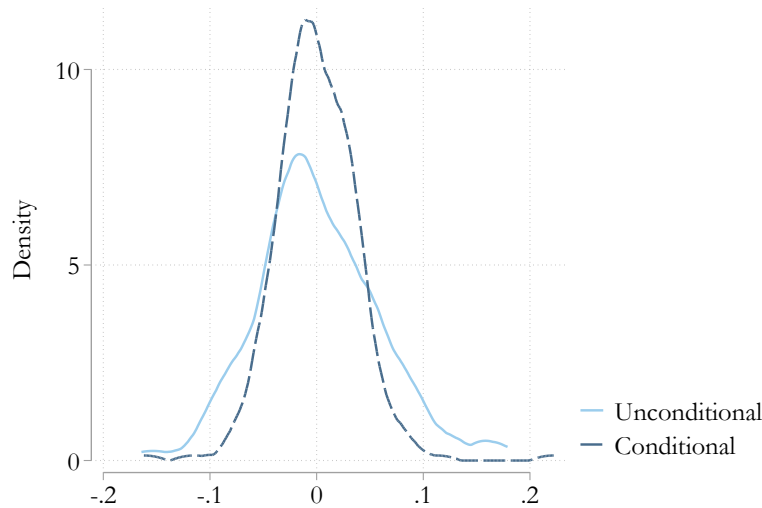
¹⁶As discussed below, our empirical strategy relies on difference-in-differences specifications controlling for patient characteristics and clinic or physician fixed effects, which absorb time-invariant level differences across clinics.

Figure 1: Average Coverage and Alert by Diagnosis Group



Note: The figure shows the share of visits that would trigger an alert against the average coverage granted for each diagnosis category in the pre-intervention period. The size of the marker is proportional to the number of visits in the pre-period in each group.

Figure 2: Distribution of Physician Coverage Propensities



Note: The figure plots the distribution of physicians' relative coverage propensities in the pre-intervention period. We use a linear shrinkage estimator following Walters (2024). We run linear regressions of coverage decision on physician fixed effects only (unconditional), and on physician fixed effects and other visit characteristics (conditional: patient age, age squared, sex, diagnosis category, month fixed effects, and day-of-week fixed effects). We drop one observation for graphic purposes.

teristics.¹⁷ Coverage propensities remain highly heterogeneous even after adjusting for observables, suggesting that physicians apply different standards to similar cases. This heterogeneity is central to the analysis that follows for two important reasons. First, it documents the agency problem that motivates the alert system: if physicians applied uniform standards, there would be no scope for an intervention designed to align coverage decisions with the insurer’s preferences. Second, the figure shows that physicians hold intrinsic preferences over coverage.

The review committees applied stricter standards for granting coverage than physicians. During the pre-intervention period, review committees overturned 4.45 percent of physician coverage decisions, 98 percent of which reversed coverage to no coverage.

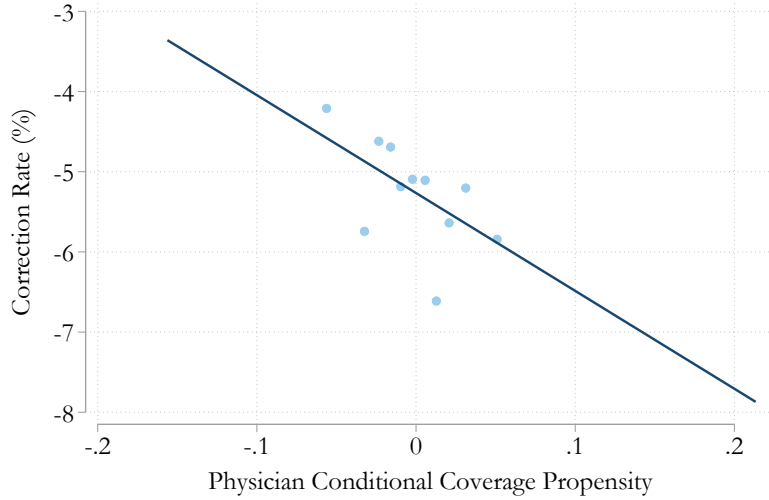
Reversal rates vary substantially across physicians. Figure 3 plots physicians’ correction rates against their conditional coverage propensity for the pre-intervention period.¹⁸ A negative slope of the correction rate against the physician’s coverage propensity reflects that physicians with higher coverage standards are reversed at a higher rate, suggesting that coverage decisions are partly explained by differing standards across physicians. Taken together with Figure 2, this finding characterizes the misalignment that the alert system is meant to address: physicians differ in their coverage standards, and those differences are systematically related to how often the principal overturns their decisions. This misalignment is the foundation on which

¹⁷We estimate linear probability models of the coverage decision on physician fixed effects, where the adjusted density also conditions on patient age and age squared, patient sex, diagnosis category, month fixed effects, and day-of-week fixed effects. We adjust the regression estimates using an Empirical Bayes shrinkage estimator following Walters (2024). The figure shows the distribution of the linear shrinkage estimated physician effects.

¹⁸We define the correction rate as the share of a physician’s covered decisions that the committee subsequently overturned: $r_j = (\text{final}_j - \text{covered}_j) / \text{covered}_j$, where covered_j is the number of cases physician j initially granted coverage, and final_j is the number remaining covered after any committee reversals. Under the null hypothesis that the committee reverses each covered case independently with a fixed probability $\bar{p}$, the expected value of r_j equals $-\bar{p}$ for every physician, regardless of how many cases she covers or how generous her coverage standard is.

the main mechanism in our model operates.

Figure 3: Reversals of Physicians’ Coverage Decisions by Physician



Note: The figure shows binned scatter plots of each physician’s “correction rate” against the conditional (initial) coverage propensities of physicians plotted in Figure 2. We define the correction rate as the share of a physician’s covered decisions that the committee subsequently overturned: $r_j = (\text{final}_j - \text{covered}_j) / \text{covered}_j$, where covered_j is the number of cases physician j initially granted coverage and final_j is the number remaining covered after any committee reversals. A negative correction rate means correction toward no coverage. We weight observations by the number of total physician cases and we include a linear fit. The figure includes pre-intervention cases only.

4 Theoretical Framework

We develop a model to highlight the key mechanisms through which the alert system may affect physicians’ coverage decisions and to derive predictions that guide our empirical analysis.

Physicians are responsible for assessing the patient’s diagnoses and determining whether to grant insurance coverage. The alert system triggers notifications for certain coded diagnoses, which physicians may or may not consider when making their decisions. A distinctive feature of this setting is that physicians can indirectly influence whether the alert is triggered by altering the coded diagnosis. To account for

this, we model the joint decision of diagnosis coding and coverage.

The model can be seen as an agency model. The insurer (the principal) has an interest in limiting coverage; the physician (the agent) holds private information about the case and retains discretion over the decision. The alert is the principal’s instrument to raise the agent’s cost of granting coverage on alert-triggering diagnoses.

The alert may affect behavior through two channels (DellaVigna and Gentzkow, 2010). The first is informational: the alert shifts the physician’s posterior assessment of the merits of the case, so that any change in behavior is one the physician would endorse on the merits.¹⁹ The second is persuasive (Stigler and Becker, 1977; Becker and Murphy, 1993): not complying is costly for physicians, changing behavior even when the alert conveys no information about the merits.

4.1 Setup

There are two possible true diagnoses, $\hat{D} \in \mathcal{D} = \{0, 1\}$. The “non-alert-triggering” diagnosis ($\hat{D} = 0$) is typically classified as work-related and does not trigger an alert, while the “alert-triggering” diagnosis ($\hat{D} = 1$) is usually not considered work-related and does trigger an alert.²⁰

Physicians observe the true diagnosis $\hat{D}$ with certainty but face uncertainty about whether the case merits coverage, represented by a latent binary variable Y . For example, physicians may know that a patient has an ankle fracture but be uncertain whether the injury is work-related and therefore covered by workers’ compensation. They decide whether to grant coverage, denoted by the binary variable C , and whether to code an alert-triggering diagnosis, denoted by the binary variable D .

¹⁹This utility could capture future rewards stemming from compensation, reputation, career benefits, etc.

²⁰In our empirical setting, alerts are only triggered in treated clinics during the post-implementation period.

We distinguish two periods: the “pre-alert period” ($t = 0$) and the “alert period” ($t = 1$). The utility from granting coverage (relative to not granting coverage) in period t , conditional on coding diagnosis D and the information set Ω_t , is defined as:

$$U_t(C = 1|D, \Omega_t) = E[Y|\Omega_t] + \lambda - \gamma \times D \times \text{Post}_t, \quad (1)$$

where the utility of not granting coverage is normalized to zero.

This specification captures several forces. Physicians assess the probability that the case merits coverage, $E[Y|\Omega_t]$, which reflects their preference for accurate decisions based on all available information. The information set Ω_t includes patient and diagnosis characteristics and may depend on the time period if the alert provides additional information.

Preferences for coverage may also depend on factors such as altruism toward patients, biased beliefs, and perceptions of monetary or non-monetary costs associated with deviating from management expectations. We capture these factors collectively in the utility function with the terms λ and $-\gamma \times D \times \text{Post}_t$. While λ denotes the pre-alert level of these factors, $-\gamma \times D \times \text{Post}_t$ reflects the *persuasive* effect of an alert, expressed as an additional cost of granting coverage when an alert is triggered ($D = 1$) in the post-alert period ($\text{Post}_t = 1$).²¹ This term captures perceived costs associated with overriding the alert’s implicit recommendation, including increases in physicians’ compliance costs with management’s preferences or liability concerns.²²

Physicians decide which diagnosis to code, and they incur a cost κ if the chosen diagnosis D differs from the true diagnosis $\hat{D}$. The joint utility of selecting coverage C and coding diagnosis D is given by:

²¹In control clinics the alert is never implemented, so $\text{Post}_t = 0$

²²Symmetrically, the absence of an alert could in principle raise the utility of granting coverage, but we find no such effect and omit it from the model for simplicity.

$$\begin{aligned}
W(C, D|\hat{D}, \Omega_t) &= C \times U_t(C = 1|D, \Omega_t) - \kappa \times \mathbb{1}[D \neq \hat{D}] \\
&= C \times (E[Y|\Omega_t] + \lambda - \gamma \times D \times \text{Post}_t) - \kappa \times \mathbb{1}[D \neq \hat{D}],
\end{aligned}$$

where κ represents the disutility from coding a diagnosis different from the true one.

Table 2 summarizes physicians' utility across possible coding choices and periods. Rows correspond to true diagnoses $\hat{D}$, columns to coded diagnoses D , and panels to pre- and post-alert periods. The term $E[Y|\Omega_t] + \lambda$ reflects assessed merits and intrinsic preferences for coverage, $-\kappa$ accounts for recoding costs, and $-\gamma$ captures the additional cost of overriding an alert.

Table 2: Payoffs Before and After Alert System Implementation

True $\downarrow$ Coded $\rightarrow$	A: Pre-Alert ($\text{Post}_t = 0$)		B: Post-Alert ($\text{Post}_t = 1$)	
	$D = 1$	$D = 0$	$D = 1$	$D = 0$
$\hat{D} = 1$	$E[Y \Omega_0] + \lambda$	$E[Y \Omega_0] + \lambda - \kappa$	$E[Y \Omega_1] + \lambda - \gamma$	$E[Y \Omega_1] + \lambda - \kappa$
$\hat{D} = 0$	$E[Y \Omega_0] + \lambda - \kappa$	$E[Y \Omega_0] + \lambda$	$E[Y \Omega_1] + \lambda - \kappa - \gamma$	$E[Y \Omega_1] + \lambda$

Note: The table shows physician's payoffs for providing coverage (relative to no coverage) depending on the coded diagnosis (in columns, D) and the true diagnosis (in rows, $\hat{D}$). Panel A corresponds to the pre-alert (control) situation while Panel B corresponds to the post-alert (treatment) situation. See text for details.

4.2 Optimal Decisions and the Effects of an Alert System

In the absence of the alert system ($\text{Post}_t = 0$), coverage and coding decisions are independent. Let $v_t \equiv E[Y|\Omega_t] + \lambda$ denote the physician's net valuation of coverage in period $t = \{0, 1\}$. Physicians maximize utility by coding the true diagnosis (i.e., choose $D = \hat{D}$) and by deciding to grant coverage based on the merits of the case and their degree of altruism, i.e., providing coverage if and only if $v_0 > 0$.

The alert system may alter physicians' decisions by providing additional information and by changing preferences in a way that links coverage and coding. We examine these mechanisms in detail below.

Changes in information. Begin with the informational effect of the alert.²³ The alert conveys information derived from the coverage decisions of a large historical sample. Under Bayesian updating, the posterior mean shifts toward the population average, with the magnitude of this shift determined by the relative precision of the physician's own signal compared to the historical average at the insurer level.

Persuasion. We distinguish between three types of cases. First, consider alert-triggering diagnoses ($\hat{D} = 1$) where $v_0 \geq 0$. Before the alert system, the physician would truthfully code ($D = 1$) and grant coverage ($C = 1$). The alert introduces an additional cost γ of granting coverage when $D = 1$. To avoid this cost, the physician may choose to code a non-alert-triggering diagnosis ($D = 0$), thereby avoiding the alert but incurring the recoding cost κ . Absent any changes in information, the net utility of granting coverage in the alert period (and optimally coding) is therefore $v_0 - \min\{\gamma, \kappa\}$. As a result, cases where physicians have strong preferences for coverage (when $v_0 > \min\{\gamma, \kappa\}$) continue to be covered, either because physicians recode to avoid the alert (when $\kappa < \gamma$) or because they ignore the alert (when $\kappa \geq \gamma$). For marginal cases where $0 < v_0 < \min\{\gamma, \kappa\}$, the introduction of the alert system causes coverage decisions to change from approval to denial.

Next, examine alert-triggering diagnoses with $v_0 < 0$. In the absence of the alert system, physicians code $D = 1$ and do not grant coverage ($C = 0$). Since the alert system imposes no additional penalty for denying coverage, physicians' behavior remains unchanged except through informational effects.

²³We present the main intuition here and provide formal derivations in Appendix E.

Finally, turn to non-alert-triggering diagnoses ($\hat{D} = 0$). Here, physicians truthfully code ($D = 0$) and grant or deny coverage based solely on case merits and altruism. The alert system does not create recoding incentives and affects coverage decisions only through information.

Overall Effect. Let

$$\Delta_d \equiv \Pr(C = 1 \mid \hat{D} = d, t = 1) - \Pr(C = 1 \mid \hat{D} = d, t = 0), \quad d \in \{0, 1\},$$

denote the change in the coverage rate within each true diagnosis type after the introduction of the alert system.

Changes in coverage decisions are due to persuasion and information. Persuasion raises the coverage threshold on alert-triggering diagnoses from 0 to $\min\{\gamma, \kappa\}$, leaving the valuation unchanged; information moves the valuation from v_0 to v_1 . Hence, we can write the change in the coverage for non-alert-triggering and alert-triggering diagnosis as

$$\begin{aligned} \Delta_1 &= - \underbrace{\Pr[0 < v_1 < \min\{\gamma, \kappa\} \mid \hat{D} = 1]}_{\text{persuasion}} + \underbrace{\Pr[v_1 > 0 \mid \hat{D} = 1] - \Pr[v_0 > 0 \mid \hat{D} = 1]}_{\text{information}}, \\ \Delta_0 &= \underbrace{\Pr[v_1 > 0 \mid \hat{D} = 0] - \Pr[v_0 > 0 \mid \hat{D} = 0]}_{\text{information}}. \end{aligned} \tag{2}$$

Appendix E derives the information terms.

Because the true diagnosis is a feature of the case and is unaffected by the intervention, the overall change in coverage is simply a weighted average of Δ_0 and Δ_1

$$\Delta \Pr(C = 1) = \Delta_0 + s_A \cdot (\Delta_1 - \Delta_0), \tag{3}$$

where $s_A \equiv \Pr(\hat{D} = 1)$ is the share of true alert-triggering cases.

Committee review. Finally, consider the review committee’s problem. Suppose the committee reverses a covered case with probability $q(v)$, weakly decreasing in the case’s merits v : $q'(v) \leq 0$.²⁴

Before implementation, an alert-triggering case is covered if $v_0 > 0$. After implementation, it is covered if $v_1 > \min\{\gamma, \kappa\}$. Let $\delta \equiv v_0 - v_1 \geq 0$ denote the downward revision in the physician’s valuation induced by the alert. The post-alert coverage rule can then be written as $v_0 > \min\{\gamma, \kappa\} + \delta$. Thus, the alert increases the valuation of the marginal covered case. Since $q'(v) \leq 0$, the expected reversal rate among covered alert-triggering cases weakly falls.

4.3 Empirical Predictions

The alert system can impact coverage decisions by changing the physician’s assessment of the merit of the case (the information channel). It also changes coverage decisions by changing the cost of covering a case (the persuasion channel). The model provides some predictions that help us test for the presence of each mechanism. Specifically:

Prediction 1: Recoding occurs only under the persuasion channel, as physicians may strategically avoid triggering the alert to avoid the additional cost γ . By contrast, information affects physicians’ assessment of coverage merits but does not change the underlying medical evaluation, leaving no incentive to alter the coded diagnosis.

Prediction 2: The extent of recoding decreases as recoding costs κ increase: Recoding is

²⁴This monotonicity assumption has an empirical counterpart in Figure 3, which shows that the committee overturns a larger share of decisions by physicians with higher coverage propensities.

worthwhile only when $\kappa < \gamma$, so cases in which switching requires a larger deviation from the true diagnosis exhibit less recoding.

Prediction 3: Under the persuasion channel, coverage falls only for alert-triggering diagnoses.

Prediction 4: Under the information channel, coverage decisions are affected for both alert-triggering and non-alert-triggering diagnoses. An alert signals low historical approval and may reduce coverage, while the absence of an alert conveys the opposite signal and may increase coverage.

Prediction 5: The information channel has a smaller effect on experienced physicians, as the weight a physician places on the alert is strictly decreasing in her own accumulated caseload (see Appendix E for details).

Prediction 6: The alert raises the threshold for granting coverage on alert-triggering diagnoses, so the cases that remain covered have higher merits on average than those covered before implementation. Fewer of them are therefore overturned by the review committee.

5 Empirical Analysis

This section evaluates the empirical predictions derived from our theoretical model. In particular, we examine whether the alert system modified physicians' behavior and assess the underlying mechanisms driving these responses.

5.1 Recoding: The Effect of the Alert System on Diagnosis Decisions

The theoretical framework predicts that the alert system may induce recoding if it affects physicians' utility through persuasion (Prediction 1). It further predicts that recoding should be more frequent when recoding costs are low (Prediction 2).

Recoding across diagnoses within a diagnosis group We first test whether the alert system influenced the probability that physicians code an alert-triggering diagnosis *within the same diagnosis group*.

Define D_{ijct} as an indicator equal to 1 if the coded diagnosis for visit i (by physician j at clinic c in week t) was an alert-triggering diagnosis. To estimate the effect of the alert intervention on coding behavior, we implement the following difference-in-differences specification:

$$D_{ijct} = \beta^W \text{Treated}_c \times \text{Post}_t + X_i' \nu + \delta_t + \rho_{d(i)} + \lambda_j + \epsilon_{ijct}, \quad (4)$$

where Treated_c is an indicator for clinics included in the alert experiment, and Post_t denotes the period after the system's implementation. The vector X_i includes patient characteristics (gender, a quadratic term in age, and industry sector fixed effects). The term δ_t comprises time fixed effects (month-year and day-of-the-week fixed effects). We also include either clinic or physician fixed effects, λ_j , depending on the specification, and diagnosis-group fixed effects, $\rho_{d(i)}$.²⁵

The coefficient of interest, β^W , captures the causal effect of the alert system on physicians' decision to code an alert-triggering diagnosis. Identification relies on

²⁵Diagnosis groups $g \in \mathcal{G}$ consist of 52 categories of diagnoses defined by the insurer. As a very small fraction of physicians works in both treated or control clinics (see Section 3), we also include a treated clinic dummy when we include physician fixed effects to preserve the diff-in-diffs interpretation of the specification.

Table 3: Recoding: Effect of the Alert System on Probability of Coding an Alert-Triggering Diagnosis

	(1)	(2)	(3)	(4)
Treated * Post	-0.022*** (0.006)	-0.021*** (0.006)	-0.017*** (0.004)	-0.010*** (0.003)
Patient Controls	No	Yes	Yes	Yes
Clinic F.E.	Yes	Yes	No	No
Physician F.E.	No	No	Yes	Yes
Diagnosis F.E.	No	No	No	Group
N	262504	262504	262504	262504
No. Clusters	82.00	82.00	82.00	82.00
Mean Dep. Var.	0.14	0.14	0.14	0.14
R-Squared	0.01	0.01	0.02	0.65

Note: Dependent variable is an indicator for an alert-triggering diagnosis coding. The table reports OLS estimates of Equation (4) as we progressively saturate the specification. All columns include month and day-of-the-week fixed effects. Patient controls comprise gender, a quadratic term in age, and industry sector fixed effects. Standard errors are clustered at the clinic level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

the parallel trends assumption, which requires no differential pre-treatment trends in coding patterns between treated and control clinics after conditioning on covariates.²⁶ All standard errors are clustered at the clinic level, consistent with the clinic being the unit of treatment (Abadie et al., 2023).

Table 3 reports OLS estimates of Equation (4) as we progressively saturate the specification. Column (1) includes only clinic, month, and day-of-the-week fixed effects, and shows that the alert system reduces the share of alert-triggering diagnoses by 2.2 percentage points. Adding patient controls in Column (2) leaves the estimate unchanged. Column (3) replaces clinic fixed effects with physician fixed effects, so that identification comes from within-physician changes in coding behavior; the estimate is -1.7 percentage points and remains significant at the one-percent

²⁶Appendix Figure A2 shows no evidence of pre-treatment differences in trends between treatment and control clinics.

level. Column (4), our preferred specification, adds diagnosis-group fixed effects and yields a coefficient of -1 percentage point, a 7 percent reduction relative to the pre-intervention baseline of 14 percent.²⁷

Heterogeneity in recoding We investigate how the intensity of recoding varies across different diagnosis groups with different pre-intervention shares of alert-triggering diagnoses. We interpret this share as reflecting the potential incentives for recoding within each diagnosis group: If the share is near zero, few diagnoses in the group trigger an alert and therefore recoding is largely unnecessary. Conversely, if the share is close to one, few non-alert-triggering diagnoses remain within the group to switch to. In such cases, recoding would require moving to a different diagnosis group, which implies a larger deviation from the truth and, arguably, higher recoding costs. According to Prediction 2, we expect less recoding in groups where such cross-group recoding would be necessary.

We classify diagnosis groups into four categories based on their pre-intervention share of visits that would trigger an alert. The first category comprises groups in the bottom seven deciles (alert-triggering share below 1 percent). The remaining three categories correspond to the 8th, 9th, and 10th deciles, which have progressively higher shares of alert-triggering visits.²⁸ We quantify the intensity of recoding within each category by re-estimating equation (4) separately for each subsample of diagnosis groups in the corresponding category.

Table 4 reports the results, showing significant recoding effects only in the two intermediate categories (Columns 2 and 3). In these diagnosis groups, the alert system

²⁷Diagnosis-group fixed effects restrict identification to within-group variation in coding, so that the estimate is not driven by shifts in the mix of conditions seen across clinics. We show below that the distribution of cases *across* diagnosis groups was not affected by the intervention, rendering the distribution of diagnosis groups exogenous to the treatment.

²⁸Appendix Figure A3 shows a histogram of the distribution of pre-treatment alert shares across diagnosis groups.

Table 4: Recoding Heterogeneity: Effect of the Alert System on Probability of Coding an Alert-Triggering Diagnosis

	Dep. Var.: Indicator for alert-triggering diagnosis coding			
	(1)	(2)	(3)	(4)
Treated * Post	0.000 (0.001)	-0.054*** (0.015)	-0.060** (0.030)	-0.000 (0.010)
Alert Share Pre.	[0,.005]	[.01,.239]	[.276,.669]	[.763,.991]
N	189,313	34,885	13,706	24,594
No. Clusters	81	81	82	80
Mean Dep. Var.	0.00	0.23	0.46	0.93
R-Squared	0.00	0.12	0.12	0.06

Note: Dependent variable is an indicator for an alert-triggering diagnosis coding. All specifications have patient controls and fixed effects for physician, diagnosis group, month, and day-of-the-week. The columns show estimation results of Equation (4) in different samples as defined by the pre-intervention alert rates of the diagnosis category (Alert Share Pre.). Standard errors clustered at the clinic level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

reduces the share of alert-triggering diagnoses by 5.4–6 percentage points, corresponding to reductions of 23.5 and 13 percent relative to baseline shares of 23 and 46 percent, respectively. These findings support Prediction 2, indicating that recoding intensity is lower when recoding entails higher costs, proxied by cases requiring a shift across diagnosis groups.

Recoding across diagnosis groups We now examine the extent to which recoding occurs *across* diagnosis groups. Cross-group recoding represents a larger deviation from the true medical condition than within-group recoding and is therefore likely to entail higher costs.

Specifically, we assess whether the monthly share of cases assigned to each diagnosis group changes with the intervention. Let $\mathcal{I}_{ct}$ denote the set of cases seen in clinic c during month t , and let $d(i)$ represent the diagnosis assigned to case $i \in \mathcal{I}_{ct}$. We compute the share of cases belonging to diagnosis group g at clinic c in month t

as

$$\text{Share}_{gct} = \frac{1}{|\mathcal{I}_{ct}|} \sum_{i \in \mathcal{I}_{ct}} \mathbb{1}[d(i) \in g],$$

and estimate the effect of the alert system on these shares using the following regression:

$$\text{Share}_{gct} = \sum_{j \in \mathcal{G}} \beta_j^R \text{Treated}_c \times \text{Post}_t \times \mathbb{1}\{j = g\} + \alpha_{cg} + \delta_{tg} + \varepsilon_{cgt}, \quad (5)$$

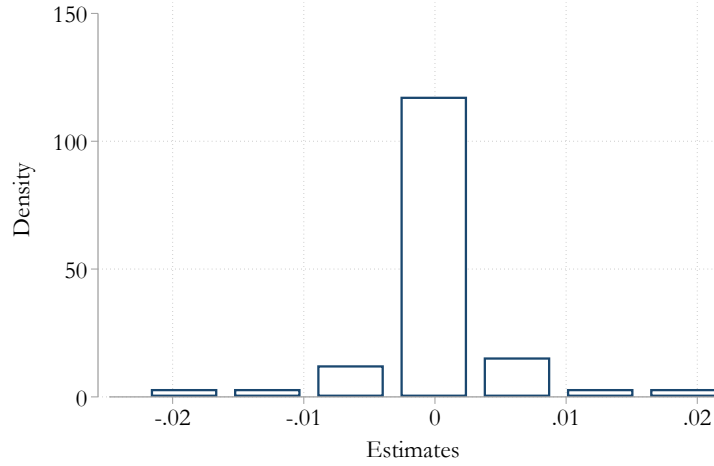
where $\mathcal{G}$ is the set of diagnosis groups, α_{cg} are a set of clinic-diagnosis group fixed effects, δ_{tg} are month-diagnosis group fixed effects and ε_{cgt} is an error term. The coefficients of interest, β_g^R , capture the impact of the alert system on the share of visits assigned to diagnosis group g . Under the null of no cross-group recoding, pre-post changes in the shares of diagnosis groups in treated clinics should be similar to those in control clinics. Thus, testing for no recoding across groups corresponds to testing the null hypothesis that all β_g^R coefficients are equal to zero.

The mean and median of the estimated β_g^R coefficients is zero and the standard deviation is 0.005. Figure 4 plots those coefficients for each of the 52 diagnosis groups. Formally, we fail to reject the null hypothesis that all coefficients are zero (p -value = 0.547), indicating no evidence of systematic recoding across diagnosis groups.

5.2 The Effect of the Alert System on Coverage Decisions

We now examine how the alert system affected coverage decisions for both alert-triggering and non-alert-triggering diagnoses. Under the persuasion channel, the alert reduces coverage for alert-triggering cases (Prediction 3); under the information channel, it alters coverage for both types of diagnoses, with stronger effects among

Figure 4: Recoding Across Diagnosis Groups



Note: The figure shows the coefficient estimates β_g^R in Equation (5). We fail to reject the joint hypothesis that these coefficients are zero, indicating absence of recoding across diagnosis groups.

less-experienced physicians (Predictions 4 and 5).

We begin by assessing the effect of the alert system on coverage using the following econometric specification:

$$C_{ijct} = \beta^A \text{Treated}_c \times \text{Post}_t \times D_{ijt} + \beta^{\text{Base}} \text{Treated}_c \times \text{Post}_t + \beta^D D_{ijt} + \nu X_i + \delta_t + \rho_{d(i)} + \lambda_j + \epsilon_{ijct}, \quad (6)$$

where C_{ijct} is an indicator for whether visit i (by physician j in clinic c at time t) receives coverage, and $D_{ijt} = 1$ if the coded diagnosis is alert-triggering.

Because physicians may recode diagnoses to avoid triggering an alert, the alert indicator D_{ijt} and its interaction with $\text{Treated}_c \times \text{Post}_t$ are endogenous. We address endogeneity with an instrumental variable approach. Specifically, for each combination of diagnosis group g and patient i 's demographic group $m(i)$, we calculate Z_{gm} as the fraction of visits that were granted coverage pre-intervention. We instrument the endogenous variables with Z_{gm} and its interaction with the treatment and post-policy

indicators.²⁹

As alert-triggering diagnoses were selected from cells with low historical coverage, Z_{gm} is a strong predictor of whether a visit triggers an alert. The Sanderson–Windmeijer conditional F -statistics in Table 5 exceed 100 for both endogenous regressors.

Three features of Z_{gm} support the exclusion restriction. First, Z_{gm} is predetermined: it is computed from visits occurring strictly before the intervention, so it cannot respond to the alert. Second, Z_{gm} is a deterministic function of (g, m) , taking the same value for any treated- and control-clinic visit in the same cell; clinic-level shocks that do not interact with (g, m) are absorbed by physician fixed effects and by the main $\text{Treated}_c \times \text{Post}_t$ effect. Third, the parameter of interest, β^A , is identified from within-physician variation in Z_{gm} across visits: two visits to the same physician on the same day can differ in Z_{gm} only because they fall in different (g, m) cells.

Conditional on the controls in Equation (6), the only channel through which the exclusion restriction can fail is post-intervention drift in the composition of (g, m) cells seen by treated clinics that is correlated with the unobserved component of the coverage decision. We fail to find evidence of such drift. Case shares across diagnosis groups are stable (Figure 4), and Appendix F shows that the alert-weighted mean of Z_{gm} does not shift within physician under $\text{Treated}_c \times \text{Post}_t$ (Table A4).

Table 5 presents the estimation results. We first present the effect of the alert on coverage *conditional* on whether the coded diagnosis triggers an alert (Columns 1–4), and then turn to the overall effect on the coverage rate in Section 5.3.

Column (1) and (2) include the alert interactions and show that the OLS esti-

²⁹We calculate the average coverage rate of the diagnosis group of $d(i)$ for the patient’s sex and age group (15–30, 31–50, 51–90) before the intervention was implemented. That is $Z_{gm} = \#\{\text{covered visits in group } g, m\} / \#\{\text{visits in group } g, m\}$, where g indexes diagnosis groups and m patient sex–age cells (15–30, 31–50, 51–90). We further instrument the interaction $\text{Treated}_c \times \text{Post}_t \times D_{ijt}$ with $\text{Treated}_c \times \text{Post}_t \times Z_{gm}$.

Table 5: Effect on the Probability of Coverage

	Conditional on alert				Overall
	OLS		IV		
	(1)	(2)	(3)	(4)	
Treated * Post * Alert	0.003 (0.016)	-0.011 (0.015)	-0.077* (0.046)	-0.130** (0.060)	
Treated * Post	-0.001 (0.004)	0.001 (0.005)	0.000 (0.006)	-0.003 (0.008)	0.003 (0.006)
Diagnosis-group F.E.	No	Group	No	Group	Group
Mean Dep. Var.	0.82	0.82	0.82	0.82	0.82
N	262504	262504	262504	262504	262504
F-Stat 1			3424.51	140.65	
F-Stat 2			2523.85	107.43	

Note: Dependent variable is an indicator for coverage. Columns (1)–(4) estimate the effect conditional on alert status and include an alert-triggering diagnosis dummy and its interaction with Treated $\times$ Post; these are the endogenous variables instrumented in the IV specifications (Columns 3–4). Column (5) reports the overall effect, estimated by OLS, omitting the alert dummy and its interaction. All specifications include patient controls and physician, month, and day-of-the-week fixed effects. Number of observations is 262,504. The F -statistics reported correspond to the Sanderson–Windmeijer (SW) conditional F -statistics for each endogenous regressor’s first stage. Standard errors are clustered at the clinic level (82 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

mates do not uncover a significant effect of the alert on coverage. Columns (3) and (4) present the 2SLS estimates, instrumenting for the alert variable and its interaction with treatment. The instruments have a strong first stage, with the F -statistics above 100. Our preferred specification in Column (4) includes diagnosis group fixed effects and shows a statistically significant reduction of 13 percentage points in the probability of giving coverage conditional on an alert-triggering diagnosis. This is a 65 percent reduction relative to the 20 percent pre-intervention coverage rate among alert-triggering diagnoses. We do not find a significant effect of the intervention on non-alert-triggering diagnoses.

The role of information

To investigate the potential role of information, we proxy the degree of physicians' ex-ante information with four physician-level characteristics measured in the pre-intervention period: tenure on the job, age, caseload (average monthly visits), and the average number of distinct days per week the physician was active. Table 6 reports 2SLS estimates of equation (6) on subsamples defined by a median split of the visits distribution along each of these four dimensions. Columns (1)–(2) split by tenure, columns (3)–(4) by age, columns (5)–(6) by caseload, and columns (7)–(8) by average days per week worked.³⁰

The results provide no support for the information channel (Prediction 5). If the alert worked primarily by updating physicians' beliefs about coverage, more experienced physicians should hold tighter priors and respond *less* to the new information; Appendix E states the comparison formally, as $\beta^A(\text{high}) - \beta^A(\text{low}) \geq 0$. We find no such attenuation. In each of the four splits the estimated effect is, if anything,

³⁰We do not detect evidence for non-linearities in three groups. The first-stage power within each subsample limits partitioning the sample into further groups.

somewhat larger in absolute value among the more experienced physicians, so the point estimates carry the sign opposite to the one the information channel predicts.³¹ The differences between the high and low groups are, however, small relative to their standard errors, and in none of the four splits can we reject equality of the two coefficients. The comparison therefore has limited power: it rules out neither the absence of a gradient nor one of the magnitude the information channel would imply. What the data do not show, even in point estimates, is the attenuation with experience that the channel requires.³²

5.3 The Overall Effect on Coverage

We estimate the overall effect of introducing the alert system on coverage by specifying:

$$C_{ijct} = \beta^{\text{Total}} \text{Treated}_c \times \text{Post}_t + \nu X_i + \delta_t + \rho_{d(i)} + \lambda_j + \epsilon_{ijct}, \quad (7)$$

where all terms are defined as in Equation (6). The coefficient on $\text{Treated}_c \times \text{Post}_t$ corresponds to the unconditional coverage effect (inclusive of any changes in coded diagnoses).

We estimate Equation (7) by OLS because no variable is endogenous to recoding as alert or its interactions are excluded.

Column (5) of Table 5 reports the results. The estimated $\hat{\beta}^{\text{Total}}$ is 0.003, with a standard error of 0.006: the alert system left the overall coverage rate essentially unchanged.

³¹The subsample estimates are less precise than the pooled effect; those for low-tenure, low-age, and high-days-per-week physicians are not statistically distinguishable from zero.

³²An alternative interpretation of these findings is that less experienced physicians may simply not know which alternative non-alert-triggering codes to use, so their muted response could reflect an inability to recode rather than the absence of a behavioral change. We test this in Appendix G. We do not find that recoding varies across any physician experience subsamples.

Table 6: Heterogeneity on Effect on Probability of Coverage

	<i>Dep. Var.: Indicator for Coverage</i>							
	By Tenure		By Age		By Caseload		By Days/Week	
	(1) Low	(2) High	(3) Low	(4) High	(5) Low	(6) High	(7) Low	(8) High
Treated * Post * Alert	-0.060 (0.055)	-0.119* (0.064)	-0.120 (0.111)	-0.133** (0.059)	-0.128** (0.062)	-0.153** (0.072)	-0.124** (0.055)	-0.150 (0.107)
Treated * Post	-0.007 (0.009)	-0.012 (0.010)	-0.008 (0.009)	-0.003 (0.009)	-0.001 (0.010)	-0.004 (0.009)	-0.006 (0.007)	-0.002 (0.013)
N	126,602	135,902	81,195	181,309	129,988	132,516	128,961	133,543
No. Clusters	76	80	68	81	81	59	80	66
No. Physicians	124	135	70	189	190	69	186	73
Range	< 34	≥ 34	< 33	≥ 33	< 79	≥ 79	< 4.1	≥ 4.1
Mean Dep. Var.	0.81	0.82	0.81	0.82	0.81	0.82	0.82	0.81
F-Stat 1	103.72	150.98	36.40	145.98	75.09	107.94	150.81	48.60
F-Stat 2	43.59	51.61	31.35	73.00	46.17	61.39	82.04	48.95

Note: Each column reports a 2SLS estimate of equation (6) on a subsample defined by a median split of a physician-level characteristic measured in the pre-intervention period: tenure on the job (columns 1–2), age (3–4), average monthly caseload (5–6), and average number of distinct days per week worked (7–8). The “Range” row reports the cutoff value, expressed in months (tenure), years (age), monthly visits (caseload), and days per week (days/week); “Low” columns are observations strictly below the cutoff and “High” columns are observations at or above it. All specifications include an alert-triggering diagnosis dummy, patient controls, and diagnosis-group, physician, month, and day-of-the-week fixed effects. The endogenous variables are the alert-triggering diagnosis dummy D and its interaction with $\text{Treated} \times \text{Post}$; the corresponding instruments are Z_{gm} and $Z_{gm} \times \text{Treated} \times \text{Post}$. The F-statistics reported correspond to the Sanderson–Windmeijer (SW) conditional F-statistics for each endogenous regressor’s first stage. Standard errors are clustered at the clinic level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We can reconcile this null effect with the sizable conditional effect of the alert estimated in Section 5.2. Equation (3) writes the aggregate change as the (alert-share) weighted sum of the coverage effects on the two true diagnosis types, whose empirical counterparts are the coefficients of Equation (6): β^{Base} on non-alert diagnoses and $\beta^{\text{Base}} + \beta^A$ on alert-triggering ones. The mapping to the estimated coefficients is not exact because $\hat{\beta}^{A,IV}$ in Column (4) of Table 5 is a LATE, defined on the visits whose alert coding responds to the instrument, whereas Equation (3) is stated as an average effect on *true* alert-triggering cases. Let this gap be θ so that

$$\underbrace{\Delta \Pr(C = 1)}_{\beta^{\text{Total}}, \text{ Column (5)}} = \underbrace{\beta^{\text{Base}}}_{\text{non-alert effect, Column (4)}} + \underbrace{s_A}_{\text{alert share}} \times \underbrace{\beta^{A,IV}}_{\text{LATE, Column (4)}} \times \underbrace{\theta}_{\text{LATE-to-ATE correction}}, \quad (8)$$

where $s_A = \Pr(\hat{D} = 1) = 0.142$ is the pre-intervention share of alert-triggering cases.

Although θ is unknown, two forces bound it below one. The first is selection of compliers. The alert list was drawn at a threshold of historical coverage, so the (g, m) cells whose coding responds to Z_{gm} are those in which coverage decisions are most contested, and in which physicians are therefore most likely to sit at the margin between granting and denying. The second is recoding. Visits that physicians recode are alert-triggering but never enter the coded-alert pool on which β^A is defined, and they retain coverage by construction. They therefore dilute the average effect on true alert-triggering cases relative to the LATE.³³ Because the alert reduced the share of alert-triggering diagnoses by seven percent (Section 5.1), this second channel alone caps θ near 0.93.

Setting $\theta = 1$ therefore provides a bound on the effect of the system on coverage.

³³Equivalently, the overall effect nets two margins: conditional on a diagnosis remaining alert-coded, physicians grant coverage less often (the intensive margin), while recoding removes some visits from alert coding altogether without changing their coverage (the extensive margin). The second margin is what drives this component of θ below one.

Evaluating Equation (8) at the Column (4) estimates gives $\hat{\beta}^{\text{Base}} + s_A \hat{\beta}^{A,IV} \simeq -0.021$, a lower bound in the value on the aggregate effect. This magnitude is not statistically different from the direct estimate in Column (5) as scaling by s_A also scales down the standard error on $\hat{\beta}^{A,IV}$.

5.4 The Effect of the Alert System on Reversals by the Review Committees

The model predicts that covered cases after the alert system was implemented have, on average, higher merits than before the alert (Prediction 6). To test this prediction, we estimate how the alert changed the rate at which the review committee reversed coverage decisions.

To see how changes in reversals help test Prediction 6, let c_0 be the coverage rate in the pre-intervention period and $m \equiv |\Delta C|$ the mass of marginal cases that the alert moved out of coverage. Partition the pre-intervention covered pool into these marginal cases and inframarginal cases that remain covered, with mass $c_0 - m$.

Let r_m and r_{inf} denote their respective reversal rates by the committee. Pre-intervention, the reversal share is $(c_0 - m)r_{\text{inf}} + mr_m$. Under the assumption that the committee's coverage standards are unchanged, only inframarginal cases remain covered after implementation, so the post-alert reversal share is $(c_0 - m)r_{\text{inf}}$. Thus, the change in the reversal share ΔO , is

$$\Delta O = -mr_m = \Delta C \cdot r_m.$$

Alternatively, we can express the pre-intervention reversal rate of marginal cases as

$$r_m = \frac{\Delta O}{\Delta C}. \tag{9}$$

Therefore, the ratio $\Delta O/\Delta C$ corresponds to the reversal rate that marginal cases would have faced had they remained covered. Prediction 6 is therefore supported if this ratio exceeds the pre-intervention reversal rate among covered cases, $r_m > r_0$. Because r_0 is a weighted average of r_m and r_{inf} , this comparison is conservative: $r_m > r_0$ implies $r_m > r_{\text{inf}}$.

We estimate ΔO using the same instrumental-variables specification as in Equation (6), replacing the coverage indicator with a reversal indicator and using the same instruments to address endogeneity from recoding.

Table 7 reports the results. Column (1) finds no significant unconditional, overall effect on reversals. Columns (2) and (3) show a significant effect of the alert on reversals of alert-triggering diagnoses when estimating the model by OLS, and Column (4) shows that the effect persists once we instrument for the endogeneity of the alert induced by recoding. Column (5), our preferred specification, adds diagnosis-group fixed effects. We find an estimated decrease of 2 percentage points.³⁴

Before the intervention, the review committee overturned $r_0 = 8.2$ percent of covered alert-triggering visits³⁵. Using Equation (9), the estimated 13 percentage point fall in coverage from Table 5 and the 2 percentage point fall in reversals of Table 7 imply $r_m = 0.020/0.130 \approx 0.15$. The marginal cases' implied reversal rate is therefore nearly twice the pre-intervention rate among covered cases, and therefore carried weaker merits than the inframarginal cases that continued to receive coverage, as Prediction 6 anticipates.

³⁴This corresponds to a 24 percent decline from the 8.2 percent pre-intervention reversal rate among physician-covered alert-triggering cases.

³⁵1.63 percent of all alert-triggering visits were covered by the physician and then reversed, and 19.9 percent were covered

Table 7: Effect on Probability of Reversal of Physician Decision

	OLS			IV	
	(1)	(2)	(3)	(4)	(5)
Treated * Post * Alert		-0.015** (0.006)	-0.014*** (0.005)	-0.019** (0.008)	-0.020** (0.008)
Treated * Post	0.004 (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)	0.005 (0.004)
Diagnosis-group F.E.	Group	No	Group	No	Group
No. Clusters	82	82	82	82	82
Mean Dep. Var.	0.045	0.045	0.045	0.045	0.045
F-Stat 1				3424.51	140.65
F-Stat 2				2523.85	107.43

Note: Dependent variable is an indicator for coverage reversal. All specifications have patient controls, physician, month, and day-of-the-week fixed effects. The endogenous variables are the alert-triggering diagnosis dummy D and its interaction with Treated $\times$ Post. Number of observations is 262,504. The F-statistics reported correspond to the Sander-son–Windmeijer (SW) conditional F-statistics for each endogenous regressor’s first stage. Standard errors clustered at the clinic level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.5 Placebo Checks

To check whether our results capture the true effect of the introduction of the alert system, we present the following placebo checks of our empirical strategy in Appendix Table A6. Panel A shows the results after changing the definition of the treatment period to July 8, 2019, to November 8, 2019, dropping the true treatment period. In Panel B we change the definition of the treatment period to April 8, 2019, to July 17, 2019, dropping the true treatment period. Finally, in Panel C we define treated clinics randomly so that the share of visits in both true and fake treated clinics is similar (≈ 45 percent). We run each of these checks for the recoding effect (Equation 4), the effect of the intervention on coverage decisions (Equation 6), and on reversals by the review committees. Table A6 shows that none of the estimated coefficients are significant in the placebo models at the 5 percent level except for a small significant effect in the treatment coefficient of the effect on coverage in Panel

C.

6 Conclusions

We examine how the deployment of data-driven information affects decision-making in the field. Our setting is one with high stakes: Primary care physicians working within workers' compensation insurance, who must decide both on a patient's diagnosis and on whether the case qualifies for coverage. We exploit the introduction of a computer-based alert system that provided physicians with historical information on coverage decisions for a subset of diagnoses. We study how this system influenced physicians' diagnosis coding and coverage choices.

We develop a theoretical model of physician decision-making where choices can be affected through both an information-based and a persuasion-based mechanism. The model delivers predictions that allow us to empirically distinguish between the two.

Several empirical findings suggest that the alert system operated through persuasion rather than information provision. First, we find evidence of strategic recoding: physicians avoided diagnoses that would trigger alerts by substituting them with similar non-triggering diagnoses. This behavior is consistent only with persuasion, since alerts should not affect diagnostic assessment. Second, responsiveness to the alert does not decline with physician experience, as an information-updating mechanism would predict. Finally, coverage changes occurred only for alert-triggering diagnoses, rather than broadly across all diagnoses, as would be expected under information updating. The alert system also improved alignment between physicians' decisions and organizational objectives, as it reduced the rate at which the coverage decisions were overturned by a secondary review committee.

Our findings have implications for the design of information systems within organizations. They indicate that computerized interventions can affect decisions through channels other than information provision. They also suggest that decision-makers may adapt their behavior in response to these interventions, particularly when doing so is not costly. Accounting for such behavioral responses is important for system design.

The persuasive effects we document matter because physicians may follow system recommendations even when these differ from their own professional judgment. This tendency may be shaped by concerns about legal liability: physicians recognize that an incorrect decision carries additional risk if it also contradicts the algorithm's recommendation. Such considerations may increase the likelihood that physicians defer to AI guidance, even in cases where it diverges from clinical judgment.

More broadly, this work contributes to understanding how decision-makers interact with data-driven information in the field. While machine learning and artificial intelligence offer potential for improving decision-making, their design needs to consider the preferences and incentives of the ultimate decision-maker. Even relatively simple alert systems, such as the one studied here, can influence decisions and have meaningful consequences, underscoring the importance of understanding the mechanisms through which they operate in real-world settings. Future research could examine how different system designs might limit strategic responses while enhancing decision quality.

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Appendix

A Additional Institutional Details

Workers' compensation insurance provides comprehensive coverage through multiple benefit types. It covers all medical expenses and provides wage replacement subsidies during periods when workers cannot perform their duties. For workers whose ability to work is permanently reduced by 15 to 40 percent, the system offers a one-time lump-sum compensation equivalent to 1.5 to 15 times the worker's salary, depending on the severity of impairment. Workers with more severe disabilities—those whose work capacity is reduced by 40 percent or more—receive ongoing pension benefits that cover their average remuneration until retirement age. In cases of worker death, the system provides survivor benefits to spouses and children.³⁶ Beyond financial coverage, workers' compensation institutions are legally obligated to implement prevention measures aimed at reducing accident rates and enhancing workplace safety standards.

Revision Committee

To ensure consistent coverage decisions among physicians, each clinic in the network operates an accident management meeting committee. These committees convene daily at noon to review all cases from the previous day and either confirm or revise initial coverage determinations. The committee comprises three key members with distinct responsibilities:

- **Chief Doctor:** Reviews the patient's event description and cross-references it

³⁶For detailed information about monetary benefits, see: <https://www.suseso.cl/613/w3-propertyvalue-63795.html> [April 18th, 2021].

with statements made to both the admission executive upon arrival and the attending physician to identify any inconsistencies. The chief doctor also analyzes the patient's clinical information and reevaluates the coverage decision made by the primary physician.

- **Commercial Manager:** Verifies that the case falls within the clinic's jurisdiction and identifies potential inconsistencies related to the accident. These findings are reported to the workplace safety expert, and this report becomes a key factor in determining the final coverage status.
- **Administrative Examiner:** Makes the final coverage decision based on the committee's discussion and the workplace safety expert's report.

B Top Alert- and Non-Alert-Triggering Diagnoses

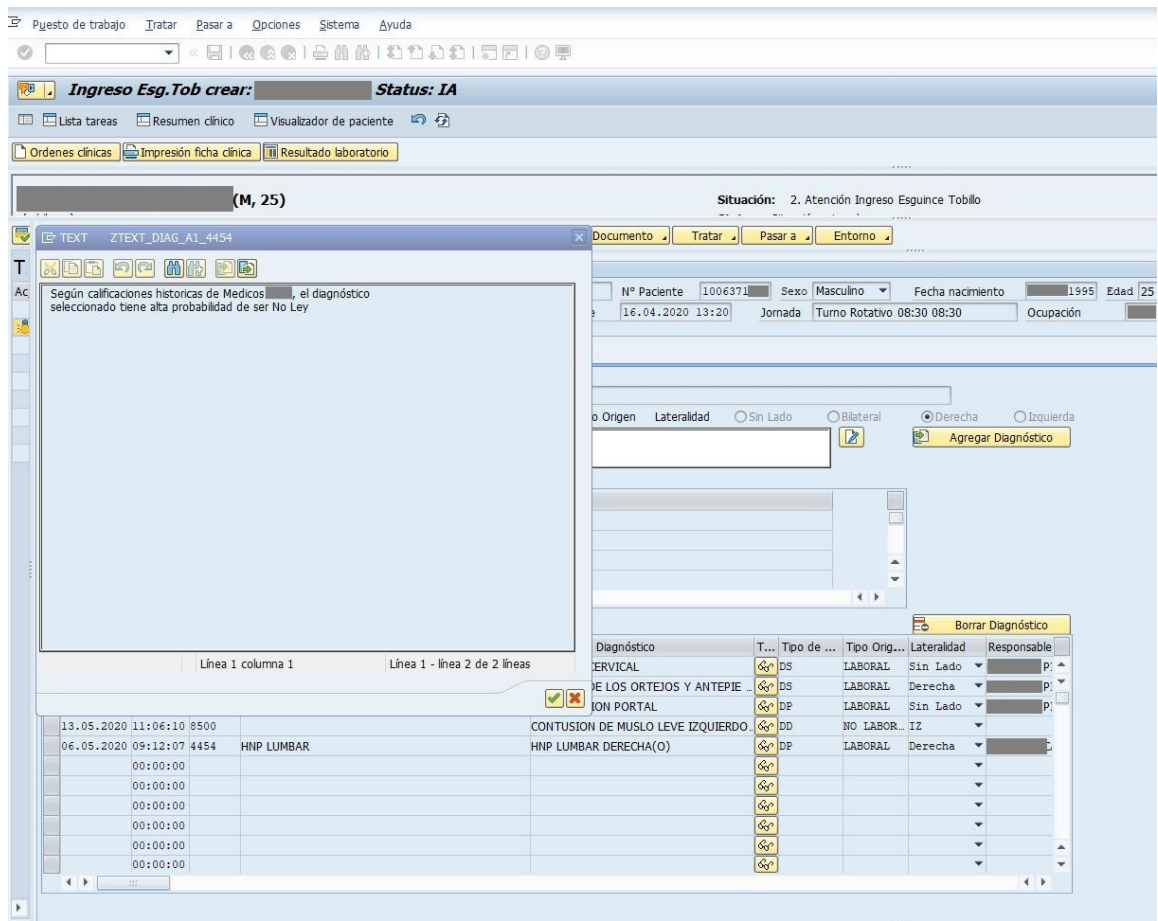
Table A1: Forty most common alert- and non-alert-triggering diagnoses

Category Code	Diagnosis	Alert-Triggering
4	Acute conjunctivitis unspecified	Yes
22	Muscular contracture	No
22	Mechanical low back pain	No
22	Lumbosciatica	Yes
22	Low back pain syndrome	Yes
23	Lateral epicondylitis	Yes
23	Plantar fasciitis	Yes
23	Synovial cyst of upper extremity	Yes
23	Bicipital tendinitis	Yes
23	Wrist extensor tendinitis	Yes
23	Wrist flexor tendinitis	Yes
23	Rotator cuff tendinopathy	Yes
23	Tenosynovitis and/or tendinitis of hand and/or wrist	Yes
24	Shoulder pain under study	Yes
24	Knee pain under study	Yes
24	Painful shoulder	Yes
25	Tietze syndrome, costochondritis	Yes
25	Facet syndrome, low back pain	Yes
29	Headache without other specification	Yes
29	Syncope or collapse (fainting, lipothymia)	Yes
30	Multiple contusions	No
31	Mild contusion of hand and/or wrist	No
31	Moderate contusion of hand and/or wrist	No
31	Corneal foreign body	No
31	Ocular foreign body (external)	No
34	Allergy unspecified	Yes
36	Biological risk accident (contact or puncture with organic fluids)	No
36	Dog bite or attack	No
37	Without medical attention	Yes
39	Simple mild head contusion, excludes TBI	No
39	Simple moderate head contusion, excludes TBI	No
46	Mild finger contusion without nail damage	No
46	Moderate finger contusion without nail damage	No
46	Finger wound with tissue loss	No
46	Mild finger wound	No
46	Moderate finger wound	No
48	Mild knee contusion	No
48	Moderate knee contusion	No
50	Grade I ankle sprain	No
50	Grade II ankle sprain	No

C Alert Screenshot

Figure A1 shows the physician’s computer interface and the pop-up alert.

Figure A1: Alert Screenshot



D Balance Analysis

In this appendix we report a balance analysis that complements that of Table 1. As noted in the main text, Table 1 collapses observations to the clinic level, which may limit statistical power. Here, we estimate visit-level or clinic-month-level versions of

Equation (4), in which pre-determined characteristics replace the X_i vector as the dependent variable, and report the coefficient on $\text{Treated}_c \times \text{Post}_t$.

We report two complementary specifications. Table A2 omits clinic fixed effects, so that the coefficient on $\text{Treated}_c \times \text{Post}_t$ is identified from the differential pre/post change in clinic composition between treated and control clinics. This mimics a diff-in-diff balance check on observables. Level differences across treated and control clinics are captured by the coefficient on Treated_c , and are absorbed by clinic and physician fixed effects in our main specifications. We find that across all five pre-determined characteristics, the $\text{Treated}_c \times \text{Post}_t$ coefficient is small in magnitude and statistically indistinguishable from zero. The Treated_c level coefficient is significant at the five percent level for physician tenure, consistent with the descriptive evidence in Table 1. These level differences are absorbed by clinic and physician fixed effects in our main specifications and are therefore not a threat to identification.

Table A3 adds clinic fixed effects. The $\text{Treated}_c \times \text{Post}_t$ coefficient captures within-clinic differential composition drift over the experimental window. As expected, absorbing clinic fixed effects sharpens the test: the coefficients on $\text{Treated}_c \times \text{Post}_t$ remain small and not statistically distinguishable from zero across all five outcomes, with standard errors that are typically smaller than in the no-fixed-effects specification.

Together, the two tables document that, conditional on the fixed effects used in our main specifications, treated and control clinics do not exhibit differential changes in patient or physician composition over the experimental window.

Table A2: Balance Analysis: Pre-determined Characteristics, No Clinic Fixed Effects

	(1) Patient age	(2) Patient female	(3) Sector HHI	(4) Phys. tenure	(5) Phys. female
Treated $\times$ Post	-0.069 (0.162)	-0.003 (0.009)	-0.014 (0.016)	0.239 (1.673)	0.006 (0.035)
Treated	-0.115 (0.221)	0.030 (0.019)	-0.037* (0.021)	-4.696** (2.242)	-0.078 (0.065)
N	262,504	262,504	1,946	260,041	262,504
No. Clusters	82	82	82	82	82
Mean Dep. Var.	39.727	0.419	0.270	24.827	0.335
Time F.E.	Month, DoW	Month, DoW	Month, DoW	Month, DoW	Month, DoW
Clinic F.E.	No	No	No	No	No
R-Squared	0.002	0.003	0.027	0.034	0.012

Note: Each column reports a separate regression of a pre-determined characteristic on $Treated_c$, $Post_t$, and $Treated_c \times Post_t$, mirroring Equation (4) with the X_i vector moved to the left-hand side. Regressions are at the visit level, except for Column (3) that is estimated at the clinic-month level and where the dependent variable is the Herfindahl-Hirschman index of industry-sector shares. Specifications include month and day-of-the-week fixed effects but no clinic or physician fixed effects. Standard errors clustered at the clinic level (82 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Balance Analysis: Pre-determined Characteristics, With Clinic Fixed Effects

	(1) Patient age	(2) Patient female	(3) Sector HHI	(4) Phys. tenure	(5) Phys. female
Treated * Post	-0.091 (0.128)	-0.002 (0.007)	-0.012 (0.017)	-0.619 (1.645)	0.026 (0.029)
N	262,504	262,504	1,946	260,041	262,504
No. Clusters	82	82	82	82	82
Mean Dep. Var.	39.727	0.419	0.270	24.827	0.335
Clinic F.E.	Yes	Yes	Yes	Yes	Yes
Time F.E.	Month, DoW	Month, DoW	Month, DoW	Month, DoW	Month, DoW
R-Squared	0.007	0.021	0.568	0.263	0.291

Note: Each column reports a separate regression of a pre-determined characteristic on $Treated_c$, $Post_t$, and $Treated_c \times Post_t$, mirroring Equation (4) with the X_i vector moved to the left-hand side. Regressions are at the visit level, except for Column (3) that is estimated at the clinic-month level and where the dependent variable is the Herfindahl-Hirschman index of industry-sector shares. Specifications include month and day-of-the-week fixed effects, and clinic fixed effects. Standard errors clustered at the clinic level (82 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Formal Derivations for the Role of Information

This appendix formalizes the information channel of Section 4.2. We derive the Bayesian shift in the physician’s posterior induced by the alert message, and use it to establish Prediction 4 and Prediction 5.

Prediction 4 states that “under the information channel, coverage decisions are affected for both alert-triggering and non-alert-triggering diagnoses. An alert signals low historical approval and may reduce coverage, while the absence of an alert conveys the opposite signal and may increase coverage.” The *Sign* paragraph below establishes this by signing the information term on each margin: on alert-triggering diagnoses the message is downward news, so the information effect is weakly negative there, and it is of the opposite sign on non-alert-triggering diagnoses. Persuasion, by contrast, operates on the alert-triggering margin alone, since γ enters the physician’s payoff multiplied by D . The non-alert margin is therefore where the two channels part company, and where the text tests this prediction.

Prediction 5 states that “the information channel has a smaller effect on experienced physicians, as the weight a physician places on the alert is strictly decreasing in her own accumulated caseload.” The *Experience gradient* paragraph below establishes this: the message corrects the physician’s prior, and the weight her posterior places on that prior, $n_0/(n_0 + N_i)$, is strictly decreasing in her own caseload and vanishes in the limit, whereas neither γ nor κ depends on N_i . This is the basis for the heterogeneity test reported in Table 6.

Consider a physician who has seen N_i cases prior to case i , of which S_i merited coverage.³⁷

³⁷Operationally, S_i counts covered cases for which the physician received no negative feedback about the coverage decision, plus rejections that later appeared mistaken. We interpret “feedback” broadly (e.g., expert panel review, subsequent signals the physician learns, etc.). Two features of this convention are worth making explicit. First, a covered case that draws no complaint is counted as

Let $p \in (0, 1)$ denote the (latent) share of cases that *merit coverage*, on which we place a $\text{Beta}(a_0, b_0)$ prior. After observing N_i cases, the posterior for p is $\text{Beta}(a_0 + S_i, b_0 + N_i - S_i)$, with mean

$$\mu_i \equiv \mathbb{E}[p \mid \text{own data}] = \frac{a_0 + S_i}{a_0 + b_0 + N_i} = \frac{n_0}{n_0 + N_i} \mu_0 + \frac{N_i}{n_0 + N_i} \hat{p}_i, \quad (\text{A1})$$

where $n_0 \equiv a_0 + b_0$ denotes the strength of the prior, $\mu_0 \equiv a_0/n_0$ its mean, and $\hat{p}_i \equiv S_i/N_i$ the physician's own empirical rate. The posterior is a weighted average of what she believed before practising and what she has since seen: an inexperienced physician is prior-dominated, an experienced one data-dominated, and $\mu_i \rightarrow \hat{p}_i$ as N_i grows.

Consider now the informative effect of the alert message given in the intervention. The alert summarizes historical decisions from many physicians. Let N_{-i} denote the number of such cases decided by physicians other than i , of which S_{-i} merited coverage, measured the same way. Treating the alert as additional observations (beyond i 's own), the posterior mean after incorporating the message is

$$\mu'_i \equiv \mathbb{E}[p \mid \text{own data} + \text{alert}] = \frac{a_0 + S_i + S_{-i}}{a_0 + b_0 + N_i + N_{-i}}. \quad (\text{A2})$$

Difference (impact of the message). Define the shorthand

$$D_2 \equiv a_0 + b_0 + N_i + N_{-i}, \quad \bar{s}_{-i} \equiv \frac{S_{-i}}{N_{-i}}.$$

meriting coverage, so S_i is a decision-based proxy for merit rather than merit itself; correspondingly, $\bar{s}_{-i}$ below is close to the historical *coverage* rate, which is precisely the quantity the alert message reports. Second, we model a single latent p rather than a diagnosis-group-specific p_{gm} , and accordingly measure experience by the physician's overall caseload rather than by her caseload within the alerted cell; our empirical proxies (tenure, age, caseload, and days worked) are overall-experience measures for the same reason.

It follows from (A1)–(A2):

$$\mu'_i - \mu_i = \frac{N_{-i}}{D_2} (\bar{s}_{-i} - \mu_i). \quad (\text{A3})$$

Thus, the alert shifts the physician’s posterior mean toward the empirical “merit-coverage” rate from other physicians, $\bar{s}_{-i}$, with weight N_{-i}/D_2 .

The weight on the message. The alert was constructed from eighteen months of decisions across the entire network, so N_{-i} is large relative both to any physician’s own caseload and to the strength of her prior. In that case $\omega_i \equiv N_{-i}/D_2 \approx 1$ and Equation (A3) reduces to $\Delta\mu_i \equiv \mu'_i - \mu_i \approx \bar{s}_{-i} - \mu_i$.³⁸ Substituting the shrinkage form in Equation (A1) and taking expectations over the physician’s own caseload, which is drawn from the same population so that $\mathbb{E}[\hat{p}_i] = p \approx \bar{s}_{-i}$ and the own-data term is mean zero,

$$\mathbb{E}[\Delta\mu_i] = \frac{n_0}{n_0 + N_i} (\bar{s}_{-i} - \mu_0). \quad (\text{A4})$$

The message moves beliefs only insofar as it corrects the *prior*, and it does so with a weight that decays in the physician’s own experience. The operative weight is $n_0/(n_0 + N_i)$ rather than ω_i : because the alert aggregates so many cases, ω_i is close to one for every physician and varies little across them, whereas n_0 is small, so $n_0/(n_0 + N_i)$ falls steeply over the range of caseloads we observe.

Coverage effects. We now map the posterior shift into the information term of the decomposition in Section 4.2. Two properties of the weight in Equation (A4) drive

³⁸The opposite case is a diagnosis for which the historical record is thin. Then N_{-i} is small, $\omega_i \approx 0$, and the message moves beliefs little: the physician knows the reported rate is itself imprecisely estimated.

what follows:

$$\frac{\partial}{\partial N_i} \frac{n_0}{n_0 + N_i} = -\frac{n_0}{(n_0 + N_i)^2} < 0, \quad \lim_{N_i \rightarrow \infty} \frac{n_0}{n_0 + N_i} = 0.$$

The weight the physician places on the message is strictly decreasing in her own experience N_i and vanishes in the limit: an arbitrarily experienced physician has already learned p from her own caseload, and the message tells her nothing.

Because the alert enters v_t only through $E[Y \mid \Omega_t]$, valuations satisfy $v_1 = v_0 + \Delta\mu_i$. Let F_d denote the c.d.f. of v_0 conditional on the true diagnosis $\hat{D} = d$, with density f_d . The information term for type d is then exactly

$$\Delta_d^{\text{info}} \equiv \Pr[v_1 > 0 \mid \hat{D} = d] - \Pr[v_0 > 0 \mid \hat{D} = d] = F_d(0) - F_d(-\Delta\mu_i) \approx f_d(0) \Delta\mu_i,$$

the last step being a first-order approximation in $\Delta\mu_i$. Three implications follow.

Sign. Because the alert list was selected on low historical coverage, the flagged cells lie below the network-wide average rate and the unflagged ones above it, so a physician holding a single prior μ_0 over p receives news of opposite sign on the two margins. On the alert-triggering margin the reported rate falls below her prior mean, $\mu_0 \geq \bar{s}_{-i}$, so by Equation (A4) the message is weakly downward news, $\mathbb{E}[\Delta\mu_i \mid \hat{D} = 1] \leq 0$, and therefore $\Delta_1^{\text{info}} \leq 0$. On the non-alert-triggering margin no message is shown, and the informational content of that silence is that the diagnosis was not among those flagged for low historical coverage. Reading the absence of an alert as such a signal places the relevant rate above her prior mean, $\mu_0 \leq \bar{s}_{-i}$, so that $\mathbb{E}[\Delta\mu_i \mid \hat{D} = 0] \geq 0$ and $\Delta_0^{\text{info}} \geq 0$: coverage weakly rises on that margin.³⁹ Persuasion, by contrast, is exactly zero on that margin: γ enters the physician's payoff multiplied

³⁹This is an assumption about how physicians interpret the absence of a message, and it is stronger than what the test requires. Because persuasion contributes exactly zero at $d = 0$, the comparison in the text has content under the weaker condition $\Delta_0^{\text{info}} \neq 0$, whatever its sign.

by D , and coding an alert-triggering diagnosis for a true non-alert case would incur both κ and γ . Taken together, the two margins are the content of Prediction 4, and the non-alert margin is where the two channels can be told apart.

Magnitude. Restricting the decomposition to $d = 1$ gives Δ_1 , the per-case effect on true alert-triggering diagnoses,

$$\Delta_1 = \Pr[v_1 > \min\{\gamma, \kappa\} \mid \widehat{D} = 1] - \Pr[v_0 > 0 \mid \widehat{D} = 1] = \underbrace{\Delta_1^{\text{info}}}_{\leq 0} - \pi,$$

which is the alert-margin term of Equation (2), where $\pi \equiv \Pr[0 < v_1 < \min\{\gamma, \kappa\} \mid \widehat{D} = 1]$ denotes its persuasion term. Because the information term is weakly negative on this margin, $|\Delta_1| \geq \pi$: the effect on true alert-triggering cases weakly exceeds what persuasion alone would deliver.

Experience gradient. Since $|\Delta_1^{\text{info}}| \approx f_1(0) \frac{n_0}{n_0 + N_i} |\bar{s}_{-i} - \mu_0|$ and this weight is strictly decreasing in N_i , the information component of the alert-triggering effect is strictly decreasing in physician experience and vanishes as $N_i \rightarrow \infty$. The gap $\bar{s}_{-i} - \mu_0$ is a primitive and does not itself vary with experience, so the monotonicity comes entirely from the prior weight. The persuasion component carries no such restriction: neither γ nor κ is a function of N_i . Prediction 5 is exactly this monotonicity. It is what renders the $d = 1$ information term testable despite its being sign-confounded with persuasion. The shift $\Delta\mu_i$ is physician-specific, so the group-level statements below are averages of the per-physician approximation over physicians in each group. For a partition into low- and high-experience physicians, holding π fixed across the two groups,

$$\beta^A(\text{high}) - \beta^A(\text{low}) \approx f_1(0) [\mathbb{E}\Delta\mu(\text{high}) - \mathbb{E}\Delta\mu(\text{low})] \geq 0,$$

because both shifts are weakly negative and the high-experience one is smaller in magnitude. Information thus predicts a *less* negative estimate among experienced physicians, and identifies the high-experience estimate as the one closest to pure persuasion. Table 6 finds the reverse ordering in the point estimates, though the differences across experience groups are not statistically significant.

F Instrumental Variable Validity and Supporting Evidence

This appendix provides additional support for the instrumental variables strategy of Section 5.2.

Equation (6) instruments D_{ijt} and $\text{Treated}_c \times \text{Post}_t \times D_{ijt}$ with Z_{gm} and $Z_{gm} \times \text{Treated}_c \times \text{Post}_t$. The exclusion restriction is that, conditional on $\rho_{d(i)}$, X_i , λ_j , δ_t , and $\text{Treated}_c \times \text{Post}_t$, the instrument Z_{gm} affects C_{ijct} only through whether the visit triggers an alert.

The construction of Z_{gm} rules out a broad class of confounders without requiring random clinic allocation. Because Z_{gm} depends only on the cell (g, m) and is computed from pre-intervention data, it is by construction uncorrelated with any post-intervention shock that is common within (g, m) . In particular, any clinic-level confounder that does not interact with (g, m) is absorbed by physician fixed effects and by the $\text{Treated}_c \times \text{Post}_t$ main effect. Identification of β^A therefore comes from within-physician variation in Z_{gm} across visits, through the triple interaction $\text{Treated}_c \times \text{Post}_t \times Z_{gm}$.

The exclusion restriction would fail if there is post-intervention drift in the composition of (g, m) cells seen by treated clinics, provided that drift is correlated with the unobserved component of the coverage decision.

We test for changes in the composition of (g, m) , by treating the instrument itself as the outcome variable and estimating

$$Z_{gm(i)} = \beta \text{Treated}_c \times \text{Post}_t + \nu X_i + \delta_t + \lambda_j + u_{ijct}, \quad (\text{A5})$$

where X_i contains patient controls (age, age squared, sex, industry sector), and δ_t

and λ_j denote time (month, day-of-the-week) and physician fixed effects. Equation (A5) is therefore a test of whether the alert-weighted distribution of Z_{gm} shifts under $\text{Treated}_c \times \text{Post}_t$.

Table A4: Effect of $\text{Treated}_c \times \text{Post}_t$ on Patient-Level Z_{gm}

	Dep. var.: Coverage pre-intervention			
	(1)	(2)	(3)	(4)
Treated * Post	-0.0004 (0.0009)	-0.0024 (0.0057)	0.0023 (0.0053)	-0.0016 (0.0040)
Alert-share range	[0.000, 0.005]	[0.010, 0.239]	[0.276, 0.669]	[0.763, 0.991]
Patient controls	Yes	Yes	Yes	Yes
Physician FE	Yes	Yes	Yes	Yes
Time FE	Month, DoW	Month, DoW	Month, DoW	Month, DoW
N	189,313	34,885	13,706	24,594
No. Clusters	81	81	82	80
Mean Dep. Var.	0.983	0.436	0.560	0.213
Adj. R Sq.	0.010	0.065	0.056	0.093

Notes: The dependent variable is $Z_{gm(i)}$, the pre-intervention coverage rate for patient i 's (diagnosis-group) $\times$ (demographic cell), where demographic cells are defined by sex and three age bins (15–30, 31–50, 51–90). Patient controls are age, age squared, sex, and industry sector. Each column restricts the sample to visits whose diagnosis group falls in the indicated bin of pre-intervention alert shares (the four bins of Table 4). The regression also include physician and time fixed effects. Standard errors in parentheses are clustered at the clinic level.

Table A4 reports estimates of Equation (A5) separately for the four pre-intervention alert-share bins of Table 4. The bins matter for interpretation. Column (1), the near-zero-alert-share bin, contains effectively no alert-triggering cells to sort away from; Column (4), the near-one-alert-share bin, contains few non-alert-triggering cells to sort toward. These two columns function as a placebo: there is no room for sorting in these groups. Columns (2) and (3) are the substantive tests—the bins in which alert-triggering and non-alert-triggering cells coexist in meaningful proportions.

The estimated coefficients in Columns (2) and (3) are statistical zeros and are small relative to the pre-period standard deviation of Z_{gm} (0.28).

G Recoding by Physician Experience

We estimate Equation (4) on subsamples defined by physician characteristics that proxy for experience and workload: tenure at the firm, age, monthly caseload, and average number of distinct workdays per week (all measured pre-intervention). For each characteristic we split the sample at the physician-level median. Table A5 reports the results. Across all eight subsamples, $\hat{\beta}^W$ ranges from -0.012 to -0.008 and is statistically significant at conventional levels in every column. Relative to subsample-specific baseline shares of 0.13–0.16, the implied reduction is between roughly 5 and 9 percent. The estimates are tightly clustered around the pooled estimate reported in Column (1) of Table 4. We find no evidence for response heterogeneity across the experience and caseload distributions.

Table A5: Heterogeneity in Recoding Across Physician Characteristics

	<i>Dep. Var.: Indicator for alert-triggering diagnosis coding</i>							
	By Tenure		By Age		By Caseload		By Days/Week	
	(1) Low	(2) High	(3) Low	(4) High	(5) Low	(6) High	(7) Low	(8) High
Treated * Post	-0.012*** (0.004)	-0.008** (0.004)	-0.009** (0.004)	-0.011*** (0.003)	-0.008*** (0.003)	-0.011** (0.004)	-0.010*** (0.003)	-0.010** (0.005)
N	126,602	135,902	81,195	181,309	129,988	132,516	128,961	133,543
No. Clusters	76	80	68	81	81	59	80	66
No. Physicians	124	135	70	189	190	69	186	73
Range	< 34	≥ 34	< 33	≥ 33	< 79	≥ 79	< 4.1	≥ 4.1
Mean Dep. Var.	0.14	0.15	0.14	0.14	0.16	0.13	0.14	0.15

Note: Dependent variable is an indicator for an alert-triggering diagnosis. Each column re-estimates Equation (4) on a physician-level subsample defined by a median split of the indicated pre-intervention characteristic: tenure (in months), age (in years), monthly caseload (visits per month), and average distinct workdays per week. Tenure, age, caseload, and days/week are computed using only pre-intervention data, fixing each physician to a single side of the split throughout the sample. Cutoff values are reported in the row labeled “Range.” All specifications include patient controls (sex, quadratic in age, industry sector), and physician, diagnosis-group, month, and day-of-the-week fixed effects. Standard errors are clustered at the clinic level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H Placebo Checks

Appendix Table A6 shows some placebo checks of our main analyses. Panel A shows the results after changing the definition of the treatment period to July 8, 2019, to November 8, 2019, dropping the true treatment period. Second, in Panel B we change the definition of the treatment period to April 8, 2019, to July 17, 2019, dropping the true treatment period. Finally, in Panel C we define treated clinics randomly so that the share of visits in both true and fake treated clinics is similar (≈ 45 percent). We run each of those checks for the recoding effect (Equation 4), the effect of the intervention on coverage decisions (Equation 6), and on reversals by the review committees. Table A6 shows that none of the estimated coefficients are significant in the placebo models at the 5 percent level except for a small significant effect in the treatment coefficient of the effect on coverage in Panel C.

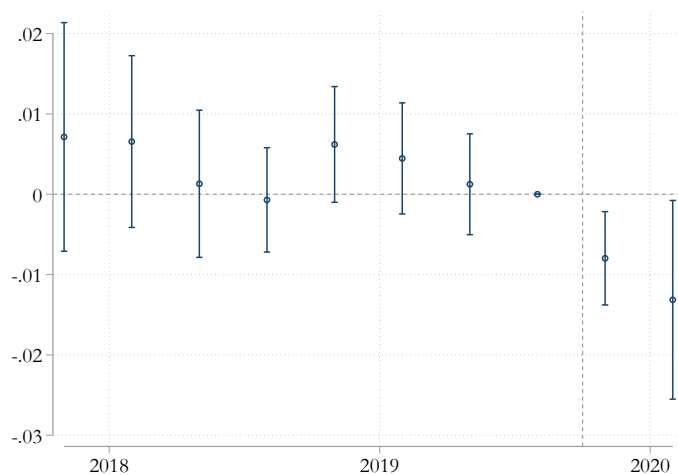
Table A6: Placebo Checks

Dependent variable:	Original			A. Fake Timing 1		
	(1) Alert	(2) Coverage	(3) Reversal	(4) Alert	(5) Coverage	(6) Reversal
Treated * Post * D		-0.130** (0.060)	-0.020** (0.008)		-0.064 (0.070)	-0.012 (0.011)
Treated * Post	-0.010*** (0.003)	-0.003 (0.008)	0.005 (0.004)	-0.001 (0.002)	0.007 (0.009)	0.006* (0.003)
N	262504	262504	262504	219785	219785	219785
Mean Dep. Var.	0.14	0.82	0.045	0.14	0.82	0.045
R-Squared	0.65	-2.09	-0.01	0.66	-2.49	-0.01
F-Stat		140.65	140.65		185.29	185.29
F-Stat		107.43	107.43		83.74	83.74
Dependent variable:	B. Fake Timing 2			C. Fake Treated Clinics		
	(7) Alert	(8) Coverage	(9) Reversal	(10) Alert	(11) Coverage	(12) Reversal
Treated * Post * D		-0.057 (0.065)	0.013 (0.008)		0.044 (0.034)	-0.005 (0.006)
Treated * Post	-0.000 (0.003)	0.000 (0.008)	-0.004* (0.002)	0.002 (0.004)	-0.014** (0.007)	-0.001 (0.004)
N	219785	219785	219785	262504	262504	262504
Mean Dep. Var.	0.14	0.82	0.045	0.14	0.82	0.045
R-Squared	0.66	-2.47	-0.01	0.65	-2.09	-0.01
F-Stat		198.69	198.69		1208.67	1208.67
F-Stat		78.85	78.85		109.10	109.10

Note: All specifications have patient controls, physician, month and day-of-the-week FE. The number of clusters is 82. Placebo checks (A)-(B) do not include the treatment period in the sample. See the text for more details. Standard errors clustered at the clinic level * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

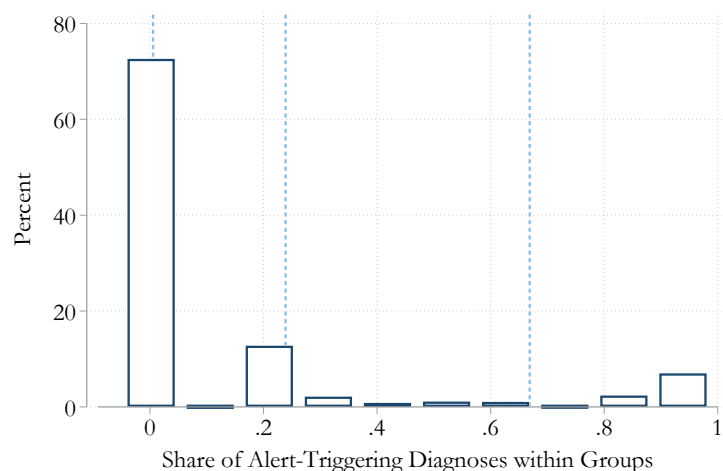
I Additional Tables and Figures

Figure A2: Dynamic Effects: Recoding



Note: The figure shows the results of estimating event-study version of Equation (4) where $Treated_c \times Post_t$ is replaced by interactions between treatment status and quarter dummies; $\sum_q \beta_q (Treated_c \times 1\{\text{quarter} = q\})$. The first quarter of 2020 includes only two weeks of February 2020.

Figure A3: Distribution of Diagnosis Groups Based on Share of Alert-Triggering Diagnoses Within Group



The figure shows a histogram of the diagnosis groups according to the share of alert-triggering diagnoses within each group. The vertical dotted lines indicate the 70, 80, and 90th percentiles of the distribution.